Andrés Aradillas Fernández

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Education

Massachusetts Institute of Technology (MIT) **Sep 2025 - Present**
PhD in Economics & Statistics *Cambridge, MA*
Major fields: Econometrics, Industrial Organization | Minor fields: Advanced Economic Theory, Computation & Statistics

Cornell University, College of Arts & Sciences **Aug 2021 – May 2025**
Bachelor of Arts with Distinction in All Subjects, Mathematics, Cum Laude *Ithaca, NY*
Cumulative GPA: 4.033/4.3 | Honors & Awards: Phi Beta Kappa ● Dean's List ● ODE Economic Honors Society

Research & Professional Experience

Research Assistant, MIT **May – Aug 2026**
Professor Victor Chernozhukov
- Developed causal inference methods for debiased machine learning under AI-learned representations.

Research Assistant, MIT **Jan 2026**
Professor Isaiah Andrews
- Studied whether bond markets price AI announcements, and constructed a macro-finance model of market responses to AI releases.

Junior Summer Analyst, Federal Reserve Bank of New York **Jun 2024 – Aug 2024**
Financial Stability team, Research & Statistics Group; research under Asani Sarkar
- Analyzed and compiled firm-level financial data for an empirical study of systemic risk across U.S. nonbank financial institutions.
- Implemented and extended MATLAB code constructing and analyzing sector-level systemic-risk measures; resulting work featured in official New York Fed publication.

Research Assistant, Econometrics Lab at Cornell University **Aug 2024 – May 2025**
Professors José Luis (Pepe) Montiel Olea, Jörg Stoye, and Chen Qiu
- Developed theoretical and computational methods for optimal decision-making and hypothesis testing for causal inference under noisy signals.
- Adapted online-learning algorithms to solve minimax problems in both econometrics and statistical decision theory.

Research Assistant, Cornell University **Oct 2022 – May 2025**
Professors Francesca Molinari and Levon Barseghyan
- Developed theoretical results for identification of choice set-dependent decision models and wrote Python simulations of discrete consumer choice with choice set-distorted preferences and choice set-dependent limited selection.

Visiting Research Intern, Columbia Business School **May 2023 – Jul 2023**
Research Assistant to Professor Jing Dong
- Constructed novel stochastic models of AI-assisted triage based on prior queueing theory work, and performed stress tests and sensitivity analysis in Python.

Additional Research Experience: Harvard Business School PRIMO Fellow (2022); Cornell Labor Dynamics Institute (2022–2023). Worked across R, Stata, MATLAB, Python, Git, synthetic control methods, and refined empirical replication methods used in top economics journals.

Selected Research

Epsilon-Minimax Solutions of Statistical Decision Problems R&R, *Journal of Econometrics*
with José Blanchet, José Luis Montiel Olea, Chen Qiu, Jörg Stoye, and Lezhi Tan
- Develop computational methods for approximately solving minimax statistical decision problems using online learning algorithms.

Pragmatic DML with AI-Learned Representations
with Victor Chernozhukov, Carlos Cinelli, Sven Klaassen, Whitney Newey, Martin Spindler, Jan Teichert-Kluge, and Suhas Vijaykumar
- Construct a causal inference framework for working with AI-learned representations that can incorporate text and image data into a broad class of estimands.

Approximate Least Favorable Distributions and Nearly Optimal Tests via Stochastic Mirror Descent
with José Blanchet, José Luis Montiel Olea, Chen Qiu, Jörg Stoye, and Lezhi Tan
- Develop stochastic optimization algorithms for large statistical decision problems using mirror descent to approximate least-favorable distributions and optimal tests.

Robust Bayes Treatment Choice with Partial Identification *Econometric Theory*
with José Luis Montiel Olea, Chen Qiu, Jörg Stoye, and Serdil Tinda
- Make precise conditions under which ex-ante and ex-post robust Bayes decision rules do not agree, and apply our results to policymakers making decisions that condition on experimental data.

Technical Skills

Programming: Python, R, MATLAB, Stata, SQL | **ML/Data:** PyTorch, scikit-learn, NumPy, pandas | **Tools:** Git, Docker, Jupyter Notebook